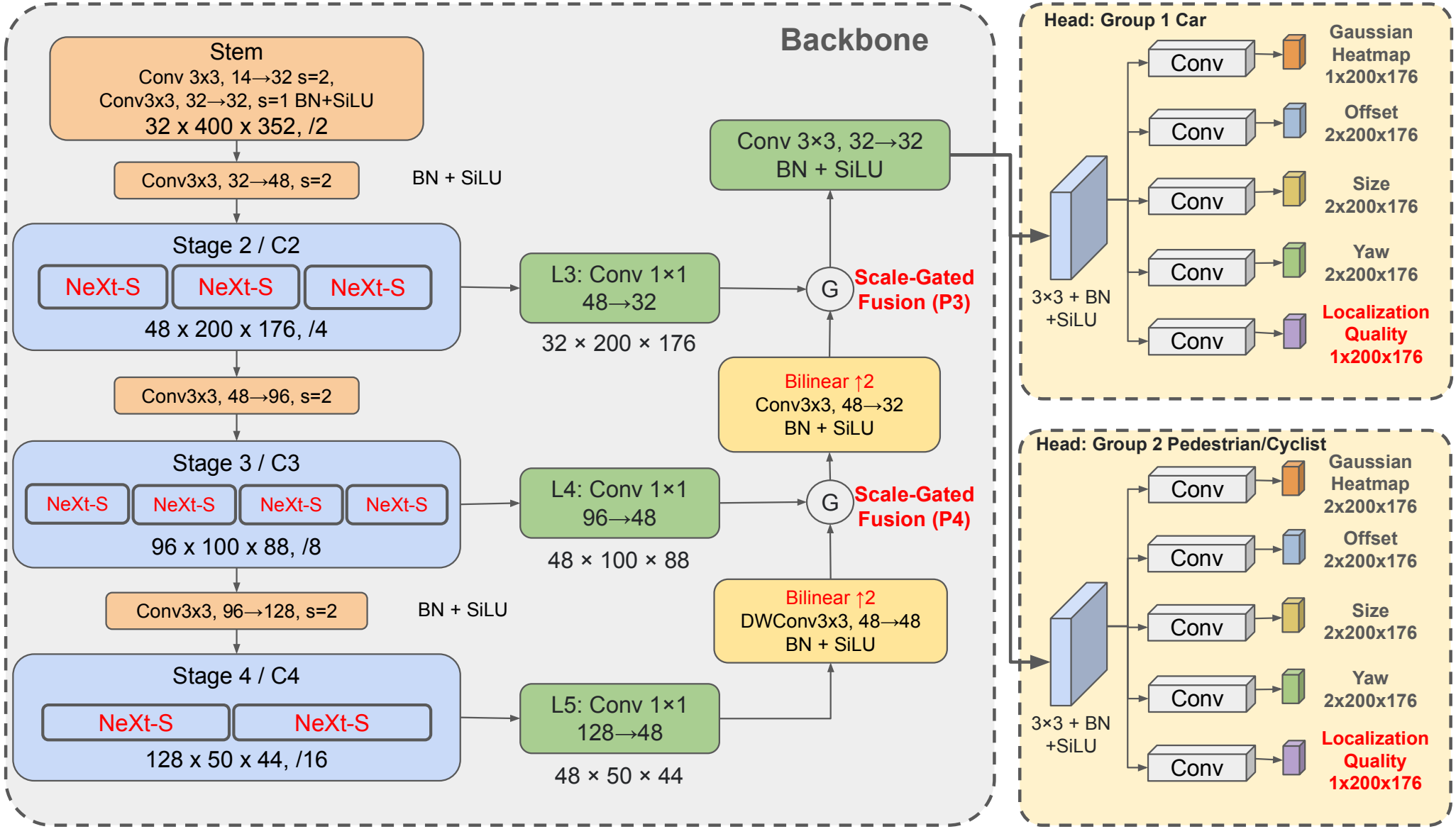
Backbone
Stem
Conv 3x3, 14→32 s=2,
Conv3x3, 32→32, s=1 BN+SiLU
32 x 400 x 352, /2
Conv3x3, 32→48, s=2
BN + SiLU
Stage 2 / C2
NeXt-S
NeXt-S
NeXt-S
48 x 200 x 176, /4
Conv3x3, 48→96, s=2
Stage 3 / C3
NeXt-S
NeXt-S
NeXt-S
NeXt-S
96 x 100 x 88, /8
Conv3x3, 96→128, s=2
BN + SiLU
Stage 4 / C4
NeXt-S
NeXt-S
128 x 50 x 44, /16
L3: Conv 1×1
48→32
32 × 200 × 176
L4: Conv 1×1
96→48
48 × 100 × 88
L5: Conv 1×1
128→48
48 × 50 × 44
Conv 3×3, 32→32
BN + SiLU
G
Scale-Gated
Fusion (P3)
Bilinear ↑2
Conv3x3, 48→32
BN + SiLU
G
Scale-Gated
Fusion (P4)
Bilinear ↑2
DWConv3x3, 48→48
BN + SiLU
Head: Group 1 Car
3×3 + BN
+SiLU
Conv
Conv
Conv
Conv
Conv
Gaussian
Heatmap
1x200x176
Offset
2x200x176
Size
2x200x176
Yaw
2x200x176
Localization
Quality
1x200x176
Head: Group 2 Pedestrian/Cyclist
3×3 + BN
+SiLU
Conv
Conv
Conv
Conv
Conv
Gaussian
Heatmap
2x200x176
Offset
2x200x176
Size
2x200x176
Yaw
2x200x176
Localization
Quality
1x200x176